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Using SHAP Analysis to Detect Areas Contributing to Diabetic Retinopathy Detection

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***Abstract*— Diabetic retinopathy (DR) is known as an important cause of blindness worldwide and serious public health concern in the population aged 20–65. With the burgeoning number of diabetes globally and its effects on patients' vision, the automatic detection of DR has received wide attention from the machine learning field. However, due to the black-box nature of deep learning and machine learning models, the interpretation and reliability of the predictions is still an issue that needs to be addressed for the successful deployment of these predictive models. In this paper, we use the SHapley Additive exPlanations (SHAP) analysis approach to detect areas of an eye image that contribute the most to the prediction of DR using transfer learning. Our predictive model achieves an accuracy of 97% and 81% for binary and multi-class classification of DR. Our SHAP analysis results show that regardless of the performance of the model, this approach can be used as a tool to interpret the prediction results with more context-sensitive information about each sample, and better understand the reasons for the classification results.**

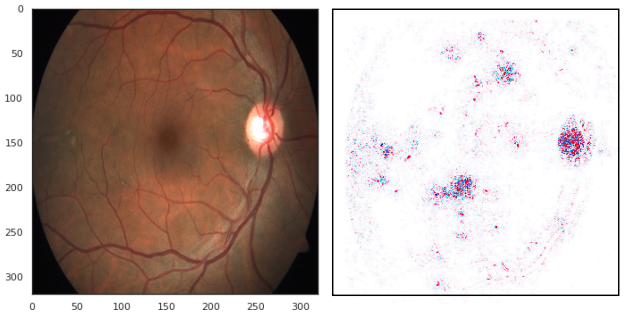
***Index Terms*— Diabetic Retinopathy, SHapley Additive exPlanations (SHAP) analysis, Deep learning.**

I. Introduction and Related Work

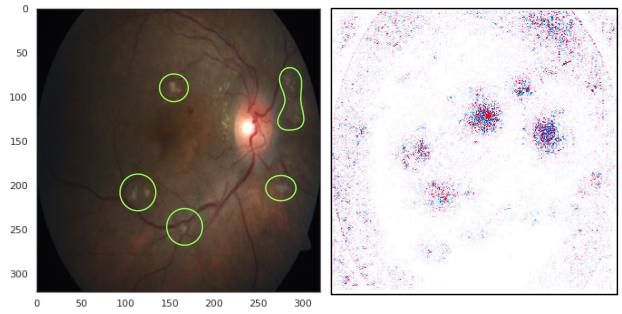
Around 9.3% (463 million) of the adult population worldwide suffered from diabetes in 2019 [1]. One of the serious complications of diabetes, called diabetic retinopathy (DR), negatively impacts blood vessels in the light-sensitive tissue of the eye (i.e. retina). In severe cases, DR can cause vision loss or blindness. Studies show that patients with long-standing diabetes (i.e. >20 years) are at risk of developing DR, with 20-30% of them ending up with blindness or visual impairment [2]. For patients with type 1 diabetes, this rate increases to more than 60%. Thus, regular screening and the early prediction of DR, especially in adults between 20-65 years old, can help mitigate the risk of blindness to a great extent [3]. At least 56% of severe DR cases can be prevented with the proper and timely detection and treatment of the disease [4]. Although the American Diabetes Association [5] recommends patients diagnosed with type 1 diabetes for more than five years undergo regular ophthalmoscopy testing, there are usually long delays in accessing screening appointments or obtaining the results, which can increase the risk of DR complications.

Due to the complexity of the detection and interpretation of DR, many general practitioners and ophthalmologists lack confidence in diagnosing the disease using fundus images [6], [7]. Thus, simplifying this task can help both the clinical team and patients for a more timely and efficient outcome. Deep learning (DL) models, including convolutional neural networks (CNN), can complement the existing efforts in practical DR detection with automatic modeling and diagnosis of different stages of DR [8]. Transfer Learning (TL) technique offers an alternative to mitigate the limitations of DL models and address the complexity of hyperparameters tuning and prevent these models from over-fitting and improve their generalizability [9].

In recent years, DR detection using DL has been addressed extensively in the literature [10]–[14]. For example, Qummar et al. [10] used the same dataset that we use in this study to detect different levels of DR severity by training an ensemble of five deep CNN models, including ResNet50, Inceptionv3, Xception, Dense121, and Dense169 to improve the features used in their deep learning architecture and increase the accuracy of their predictions for different stages of DR. The new architecture proposed by their study outperformed the existing models with more accurate predictions of stage-based DR. Similarly, Khalifa et al. [11] used different deep transfer learning models, including AlexNet, Res-Net18, SqueezeNet, GoogleNet, VGG16, and VGG19 to detect DR and found



(a) The original image and the SHAP results for DR=0



(b) The original image and the SHAP results for DR=1

Fig. 1: Sample images presenting the areas of eye used for detecting the DR status in our binary model

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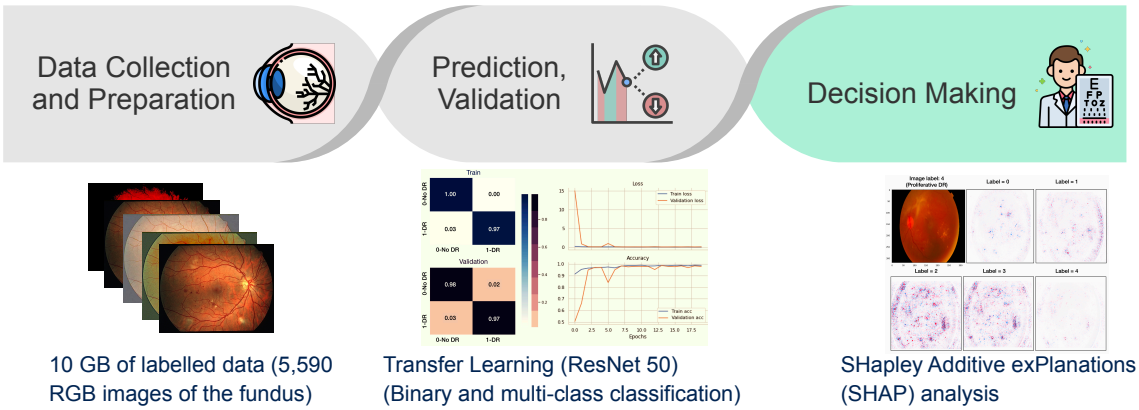


Fig. 2: The overall process of this study, including the main stages and the data and artifacts associated with each stage. This process is implemented sequentially and validated through different iterations.

that the model trained based on AlexNet outperformed the other pre-trained models with 97.9% accuracy. To compare the outcomes of the research on DR detection using machine learning, Pak et al. [12] conducted a comparative study on the two widely used conventional architectures (i.e. DenseNet, ResNet) and the recent optimized model (i.e. EfficientNet). The results of their study show that EfficientNet outperforms the other models in terms of accuracy and the number of parameters used for the model architecture. While these studies addressed the challenging problem of DR detection with well-performing predictive models, the interpretation of the outcomes of these models in the context of early DR detection, which can assist clinicians with their diagnoses, has not received enough attention.

To address this limitation, in this paper, we use SHapley Additive exPlanations (SHAP) analysis to explain the results of a deep transfer learning model for detecting either DR as a binary problem or different stages of DR severity. The results of our study show that leveraging the contextual data provided by SHAP analysis can help clinicians when interpreting machine learning decisions and provide more information about individual decisions (predictions) of the model (e.g. see Figure 1). We also show that this approach can be utilized to only focus on a subset of features that contribute the most to the prediction results.

The rest of this paper is organized as follows. The details of the dataset used for this study, the prediction models, and the analysis approach to detect the contributing features are presented in Section II. Section III presents the key findings of this study, followed by discussions of the results. Finally, we conclude this paper in Section IV areas to explore as next steps.

II. Methods and Materials

Figure 2 illustrates the overall process of this study and highlights the primary goal of this study (i.e. assisting domain experts with their decisions when using machine learning and prediction models). More specifically, this section outlines the dataset we used to train and validate our models and the data preparation steps we followed before the prediction task. We also present and discuss the prediction model used in this study, followed by the description of the SHAP technique used to complement and interpret our results.

A. Dataset and Data Preparation

To train and test our models, we used the Asia Pacific Tele-Ophthalmology Society (APTOS) 2019 dataset [15]. The dataset of fundus images of high quality in India gathered from Aravind Eye Hospital, which consists of about 10 GB of labelled data, including 5,590 RGB images of the fundus. The training dataset includes 3,662 images labeled with five categories: 1. No DR-0 (n=1,805), 2. Mild-1 (n=999), 3. Moderate-2 (n=370), 4. Severe-3 (n=295), and 5. Proliferative DR-4 (n=193).

To expand the size of our training set to improve the performance and generalizability of our predictive model, we used the ImageDataGenerator library in Keras for image augmentation. To do this, we used image rotation and vertical and horizontal flipping to simulate possible deviations of eye orientations. Moreover, given that the provided images in the APTOS dataset are in RGB format, we rescaled the images with a 1/255 factor (i.e. range 0-1) to make the learning process more efficient.

B. Prediction Model

To predict DR using the APTOS dataset, we used the Residual Networks with 50 layers (ResNet50) [16], [17] as the first step of our transfer learning model. This base model is followed by a GlobalAveragePooling2D layer, a dropout layer with a rate of 0.5, a dense layer of size 2,048 and another dropout layer before the output layer. We used the `categorical_crossentropy` loss function and `relu` activation for the dropout layers and the `softmax` activation for the output layer. We trained and tested our models using both binary (DR=0, DR=1) and multi-class (DR$\in \{0, 1, 2, 3, 4\}$) datasets. For the multi-class classification, as the dataset is highly unbalanced, we considered

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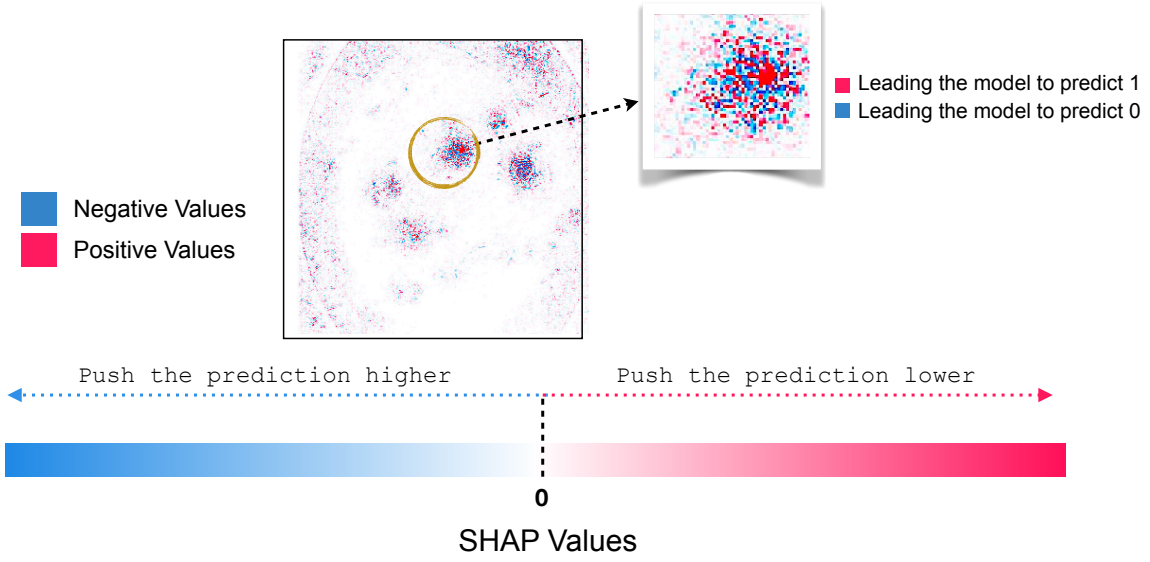


Fig. 3: Interpretation of SHAP values. The sample SHAP output presents a fundus image with label DR=1.

higher weights for the minority class when calculating the class weight during the training phase [18]. Moreover, to ensure the model's performance is not due to over-fitting, we plotted the loss and accuracy values across all epochs. At the same time, we also set EarlyStopping to avoid over-fitting in our learning process. The computations and the deep learning model for both binary and multi-class classification tasks were implemented using Python 3.7 with TensorFlow 2.0 [19], Keras [20], and Scikit-learn [21] using an Nvidia Tesla P100 GPU with 12GB of device memory.

C. SHAP Analysis

To explore the features that contributed to the outcomes of our prediction models, we used the SHapley Additive exPlanations (SHAP) analysis [22], [23] technique. SHAP is a method adopted in game theory that measures each player's contribution to a collaborative game. In the image processing and general machine learning context, each SHAP value determines how much each feature (i.e., area of an image) contributes to the target prediction in both positive and negative ways. While the existing feature analysis techniques calculate the importance of features globally, the SHAP approach calculates the local feature importance for each dataset image and assigns each feature an importance value for a specific prediction. This approach can address the inconsistency problems in existing feature importance techniques and mitigate the effect of misinterpretations associated with these inconsistencies [24]. In this study, we applied the SHAP analysis technique to highlight the locations of fundus images that mostly show either the presence of DR or different DR stages in terms of the severity of the disease.

III. Results and Discussions

In this section, we present the results of our predictive model and elaborate on the efficacy of using SHAP analysis

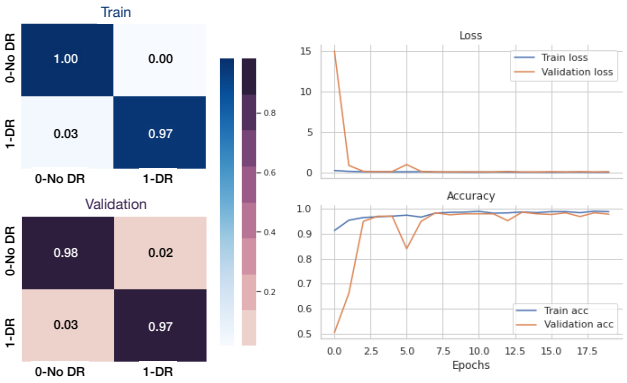


Fig. 4: Confusion matrix presenting the results of binary classification using the deep learning model

in explaining these results. Figure 3 provides an illustrative example of the interpretation of SHAP values that we use to discuss the results of our study in this section. We also list the main limitations that could potentially impact the understandinginterpretation of the results of this study, along with brief discussions about our mitigation strategies.

A. Key Findings

Figure 1 presents samples results of using SHAP analysis for detecting contributing areas to the DR detection problem in our study. In this figure, we presented two samples of data that are similar in terms of presentation and configuration of the original image, but one is labeled as DR=0 (Figure 1a), and the other one is a DR image (Figure 1b). The SHAP output presented to the right side of each image highlights the areas our deep learning model used for the prediction. Red pixels increase the model prediction, while blue pixels

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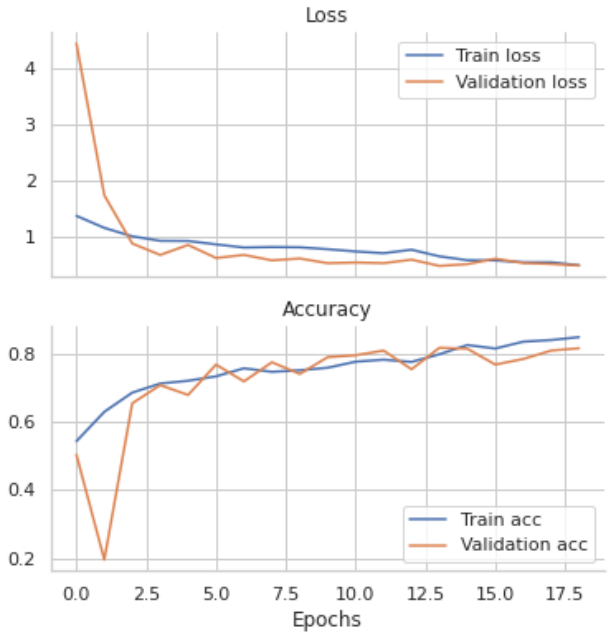


Fig. 5: The loss and accuracy diagrams for the multi-class classification task shows no over-fitting in the learning process

decrease the prediction. The green areas in figure1 represent the areas of eye with lesion identified manually by a medical expert from Alberta Ocular Barchytherapy Program. Interestingly, we can see from this figure that areas that are highlighted in red in the no-DR labeled image (Figure 1a) are highlighted in blue in the DR labeled image (Figure 1b). This can be used to define a map of contributing features by highlighting areas to assist clinicians in diagnosing DR by looking at fundus images.

Figure 4 presents the confusion matrices for the training and validation tasks of our binary classifier. The loss and accuracy plots for the training and validation sets in this figure show our model's generalizability and that the model's high performance (i.e. accuracy: 98%, loss: 0.1) is not due to over-fitting. We also evaluated the performance of our multi-class predictive model labels for different stages of DR (listed in Section II-A). We achieved an overall validation accuracy of 81% and loss of 0.48 for this task (see Figure 5). The model yielded its best performance for detecting the **no-DR** class with an accuracy of 98%, followed by **Moderate DR** with an accuracy of 71%. Each of the **Mild DR** and **Sever DR** classes were detected with 66% and 61% of accuracy, respectively. The **Proliferative DR** category comes last, as the model could only detect 45% of these samples correctly. The majority of misclassified samples from this class were assigned to the Mild- or Severe DR categories, which could be because of the small size of this class or the clinical similarities between the DR signs in these three categories. Our detailed prediction results also show that most False Negatives (FNs) of the Severe-DR class are assigned to the Moderate-DR category (i.e. 39%). This implies a clear need for more investigation of the data or the learning architecture when developing automatic techniques for detecting progressive DR.

Another key finding of this study is that, for prospective DR detection using predictive models, regardless of the quality of labels and the richness of the pre-trained models (e.g. ResNet50 in our study), it is important to perform feature engineering before passing the data through a deep learning model. For example, as illustrated in Figure 6, for our multi-class classification task, a **Proliferative DR** (i.e., label = 4) image is misclassified to the moderate DR (i.e., label = 2) due to the lack of context information in the model for the Proliferative DR category. The highlighted areas in the SHAP outputs present data points that influenced the model's decision-making about the DR outcome. From the example provided in Figure 6, we can see that the model had the slightest knowledge about the correct category. This can be mitigated by adding more samples to the Proliferative DR category along with performing a more customized feature engineering (e.g. SHAP analysis in this case). This figure also confirms our earlier discussion about the similarity of the Moderate- and Severe-DR (i.e., label = 2 and 3) categories. We can see that almost the same areas are highlighted in both categories as the most influential areas. Moreover, the results of our SHAP analysis tasks are in line with a recent study on using deep learning to predict evolving DR in individual patients [25]. This study also confirms the application of the SHAP technique in interpreting deep learning models for detecting progressive DR.

B. Limitations

Several limitations influence the generalizability of this study's results. First, we only used one deep learning architecture to explore the utilization of SHAP analysis for interpreting the outcomes of the DR detection problem. Using different models on the same dataset might generate different results for the SHAP approach. However, the large size of this dataset and the consistency of results in our study mitigates the risk of this limitation to a great extent. Due to limited computation power, we configured our deep learning model with `epoch<30`, which could impact the learnability of our model. However, the high performance of the model in the binary classification task and the loss and accuracy plots (Figure 4) show that the greater epoch may not significantly change the results of our study.

IV. Conclusion and Research Implications

The interpretation of deep learning models in patient outcome predictions can bridge the gap between data scientists and clinicians. However, despite the power of deep learning techniques in modeling the complexity of image datasets, the black-box nature of these models may question their usability when it comes to real-world decision-making tasks. In this study, we used SHAP analysis to provide more context about each image's features that contribute to a specific prediction (i.e. DR detection in our study) [26]. The results of our study

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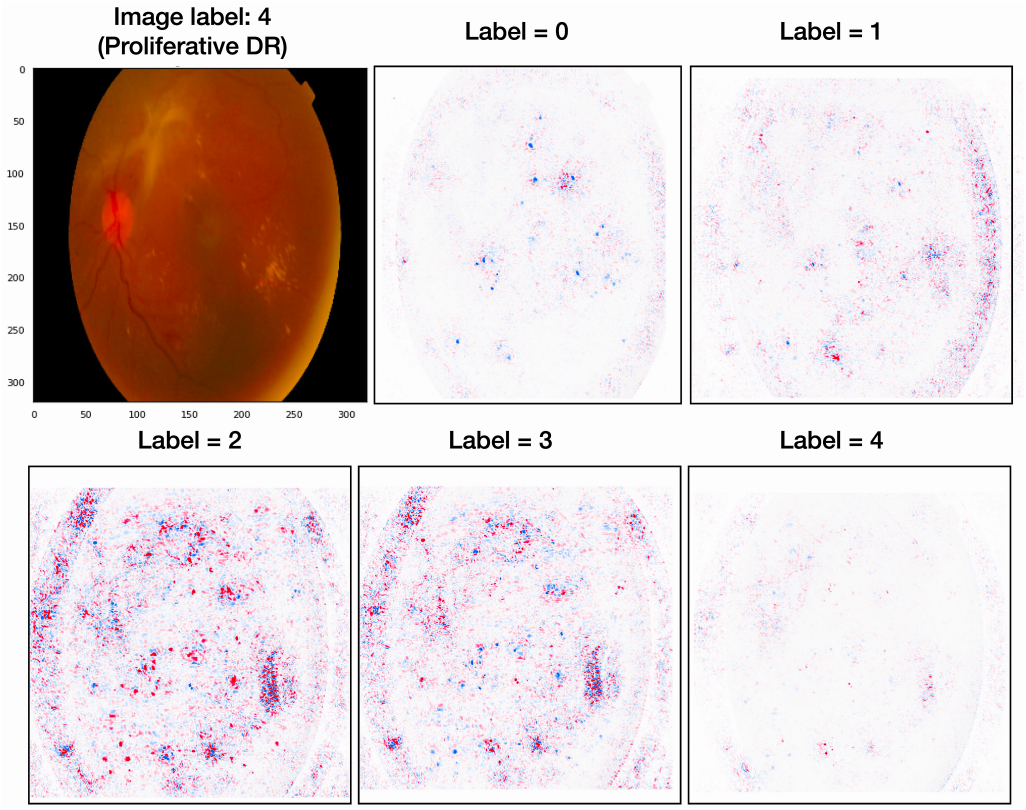


Fig. 6: A sample misclassified image from the Proliferative DR category. The SHAP analysis result can help physicians interpret why this sample is misclassified to moderated DR (Label = 2).

show that SHAP analysis consistently detects the areas that can differentiate between DR and no-DR samples (example: Figure 1). One of the main research implications of this study is validating our results by involving clinicians in the interpretation phase and ensuring the SHAP approach can detect areas of concern in image data for DR patients. Also, to evaluate the generalizability of this approach, we are planning to replicate the presented study for Uveal Melanoma detection by using a Canadian dataset of patients diagnosed with Uveal melanoma. The results of this follow-up study will help further evaluate the SHAP approach's efficacy.

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